

Preclass Learning: APIs P1 - HTTP Requests and JSON (ICA 13)

1. Application Programming Interfaces (APIs)

An API defines the set of rules that governs how two pieces of software communicate. A Web API allows a client (your R/Python script) to communicate with a server (a data source).

- **Endpoint:** A specific URL that represents a data resource on the server (e.g., `/api/v1/users`).
- **Parameters:** Variables passed to the endpoint, often in the query string (`?key=value`), to filter or modify the request.

2. HTTP Methods

The HyperText Transfer Protocol (HTTP) defines the actions a client can perform:

- **GET: The primary method for data retrieval.** Requests a representation of the specified resource. This method should only retrieve data and have no other effect.
- **POST:** Used to submit data to the specified resource, often resulting in a change in state or a new resource being created (e.g., submitting a comment).
- **PUT/DELETE:** Used for updating or deleting resources, respectively.

3. JSON Data Format

- **JSON (JavaScript Object Notation):** The most common data format for APIs. It is human-readable, lightweight, and easily translated into standard data structures (lists and data frames/dictionaries).
 - **Structure:** Uses key-value pairs (`{"key": "value"}`) and arrays (`["item1", "item2"]`) to represent hierarchical data.
- **Parsing:** The process of converting the raw JSON text string into a native programming object (like an R list or data frame) that can be easily manipulated.